

# Deep Reinforcement Learning

## *Optimal Control & Feedback Control*

Prof. Dr. Sebastian Peitz

Chair of Safe Autonomous Systems, TU Dortmund

Summer term 2026

# Content

# Content

- Recap of model-free RL
- Open-loop optimal control
  - Trajectory planning
  - Solution approaches
  - Stochastic optimal control
  - The linear case
- Closed-loop control
  - Model Predictive Control (MPC)
  - Differentiable Predictive Control (DPC)

# Where are we?

| Chapter | Topic                                                        | Content                                  |
|---------|--------------------------------------------------------------|------------------------------------------|
|         | <b>Basics &amp; tabular methods</b>                          |                                          |
| 1-5     | Bandits, MDPs, Dynamic Programming, Monte Carlo, TD Learning | RL basics in finite dimensions           |
|         | <b>Deep-learning-based methods</b>                           |                                          |
| 6-13    | DQN, policy gradients, actor-critic, PPO/SAC, exploration    | Deep RL from basics to modern algorithms |
|         | <b>Model-Based Control</b>                                   |                                          |
| 14      | Optimal control & feedback control                           | How to do control when we know the model |
|         | Model-based reinforcement learning                           |                                          |
|         | <b>Advanced Topics</b>                                       |                                          |

*Lecture contents*

# Recap: model-based and model-free RL

- Until now, knowledge of a model only in the simplest RL approaches.
  - Dynamic Programming for tabular RL, e.g., policy or value iteration.
  - We need to know  $p(s'|s, a)$  to update our value estimate!
- All the advanced algorithms: **model-free RL**.
  - Collect experience by following the current policy  $\pi$ ,
  - then update our estimate of  $V$ ,  $Q$ ,  $A$  or  $\pi$  only from the data.

## Advantages / disadvantages of model-free RL

- + No model knowledge required. All we need is access to the system / a black-box simulator.
- + No bias due to inaccurate model.

Today: Can we do better if we know the dynamics?

We often know the dynamics:

- Games (e.g., Chess, Go, Atari games).
- Easily modeled systems (e.g., car navigation).

**Algorithm: Value iteration for estimating  $\pi \approx \pi^*$ .**

**while** ( $\Delta > \theta$ ):

$\Delta \leftarrow 0$

**for**  $s \in \mathcal{S}$ :

$V_{\text{old}}(s) \leftarrow V(s)$

$V(s) \leftarrow \max_{a \in \mathcal{A}} \sum_{s' \in \mathcal{S}} p(s'|s, a) [r + \gamma V(s')]$

$\Delta \leftarrow \max(\Delta, |V_{\text{old}}(s) - V(s)|)$

+ Easier to implement.

— Data-hungry! Can be very sample-inefficient.

— Ignorant of prior knowledge.

Also, we can often learn the dynamics:

- System identification: fit unknown parameters of a known model.
- Learning: fit a general purpose model to observed transition data.

- Simulated environments (e.g., robotics, video games)

# What is a model?

- Very generally, a **model** is an MDP tuple  $(\mathcal{S}, \mathcal{A}, p, r, \gamma)$ , just as we have seen before. Most notably, we have
  - the dynamics model  $p(s'|s, a)$ ,
  - the reward probability  $r \sim p(\cdot|s, a)$ .
- The state and action spaces  $\mathcal{S}$  and  $\mathcal{A}$  are usually assumed to be known.
- The discount factor is given or can be chosen by an expert.

## Types of models

### 1. Probabilistic vs. deterministic

- probabilistic:

$$s' \sim p(\cdot|s, a), \quad r \sim p(\cdot|s, a).$$

- deterministic (a special case of the former):

$$s' = f(s, a), \quad r = \ell(s, a).$$

### 2. Known vs. unknown

- **True MDP:**  $p$  and  $r$  (or  $f$  and  $\ell$ ) are perfectly known a priori.
- **Approximated MDP:**  $p$  and  $r$  (or  $f$  and  $\ell$ ) need to be learned,

$$\tilde{p} \approx p, \tilde{r} \approx r \quad \text{or} \quad \tilde{f} \approx f, \tilde{\ell} \approx \ell.$$

# Open-loop optimal control



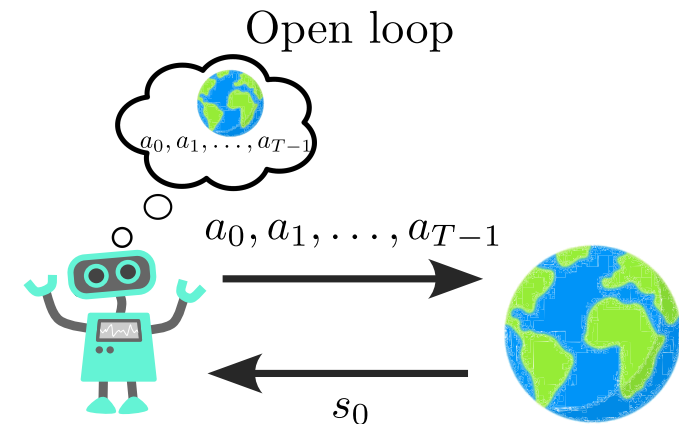
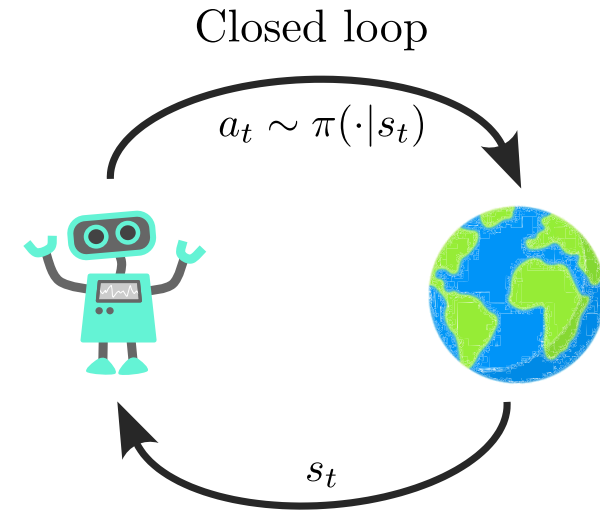
# Planning & optimal control

- For now, we assume that the model
  - is known,
  - is fully deterministic.
- In the episodic case ( $T < \infty$ ), the RL problem becomes

$$\min_{a_0, \dots, a_{T-1}} \sum_{t=0}^T \ell(s_t, a_t) = \min_{a_0, \dots, a_{T-1}} L(a) \quad (1)$$

subject to  $s_{t+1} = f(s_t, a_t), t = 0, 1, \dots, T-1$ .

- Eq. (1) is referred to as an **optimal control problem (OCP)**.
  - $\ell$  is the *stage cost*.
    - 💡 Closely related to the negative reward  $-r$ !
  - $L$  is the *cost* that we seek to minimize.
    - 💡 Closely related to value  $V$ !
  - Optimal control is usually formulated as a *minimization problem*, e.g., the distance to a desired trajectory.
- Generally, optimal control refers to the **open-loop / planning** setting.



Only measure at  $s_0$ , no feedback

Inspired by Sergey Levine's CS285 lecture.



# The deterministic vs. the stochastic case

- In RL, we usually assume that we have a stochastic problem for various reasons:
  - stochastic dynamics,
  - stochastic actions/policies,
  - stochastic rewards.
- In the open-loop setting, stochastic optimal control is suboptimal!  
⇒ Why?

## Stochastic dynamics

$$p(s_0, \dots, s_T | a_0, \dots, a_{T-1}) = p(s_0) \prod_{t=0}^{T-1} p(s_{t+1} | s_t, a_t).$$

- The optimal action  $a_1$  depends on  $s_1$ , which we cannot know for certain at  $t = 0$ .
- The same is true for all following states and actions.

## Some approaches for stochastic optimal control

- Certainty Equivalence (CE)
  - Simply replace all stochastic quantities by their expectations/mean values.  
+ Computationally cheap.  
— Completely ignores variance.
- Mean-variance / risk-sensitive optimization
  - define loss function composed of mean and variance:

$$\min_a \mathbb{E}[L] + \gamma \text{var}[L].$$

- + Accounts for variance.
- Need additional compute to estimate  $\text{var}[L]$ .

- Sample-path optimization
  - Monte-Carlo estimate of expected loss by multiple simulations with different noise realizations  $\omega_i$ :

$$\min_a \frac{1}{N} \sum_{i=1}^N L(a, \omega_i).$$

+ Can estimate arbitrary distributions (beyond Guassian).

— Compute grows with  $N$ .

# Solving the OCP: single shooting

$$\min_{a_0, \dots, a_{T-1}} \underbrace{\sum_{t=0}^T \ell(s_t, a_t)}_{=L(a)} \quad \text{subject to} \quad s_{t+1} = f(s_t, a_t), \quad t = 0, 1, \dots, T-1. \quad (1)$$

A very common approach to solve the OCP (1) is via **single shooting**, where assume that the states  $s_t$  are implicitly defined by choosing the actions  $a_t$ .

💡 This holds for most systems of interest. The basis for this are existence-and-uniqueness theorems, e.g., by **Picard-Lindelöf**.

## Single shooting

*Initialize:* Initial state  $s_0$ , action sequence  $a^{(0)} = (a_0^{(0)}, \dots, a_{T-1}^{(0)})$ .

**for**  $j = 0, 1, 2, \dots$ :

    simulate the system following  $a^{(j)}$  to get  $s^{(j)} = (s_1^{(j)}, \dots, s_T^{(j)})$

    assess the performance by computing  $L(a^{(j)})$

    gradient descent to update our actions:

$$a^{(j+1)} = a^{(j)} - \eta \nabla_a L(a^{(j)})$$

**Remaining question:**

How do we compute the gradient

$$\nabla_a L(a^{(j)})?$$

# Gradients in single shooting (1)

$$\min_{a_0, \dots, a_{T-1}} \underbrace{\sum_{t=0}^T \ell(s_t, a_t)}_{=L(a)} \quad \text{subject to} \quad s_{t+1} = f(s_t, a_t), \quad t = 0, 1, \dots, T-1. \quad (1)$$

The procedure of computing the gradient is closely related to the backpropagation algorithm!

1. Turn the constraint into an objective via the Lagrangian formalism, where  $\lambda$  is the adjoint state:

$$\begin{aligned} \mathcal{L}(s, a, \lambda) &= \sum_{t=0}^T \ell(s_t, a_t) + \sum_{t=0}^{T-1} \lambda_{t+1}^\top (f(s_t, a_t) - s_{t+1}) \\ &= [\ell(s_0, a_0) + \lambda_1^\top f(s_0, a_0)] + [\ell(s_1, a_1) + \lambda_2^\top f(s_1, a_1) - \lambda_1^\top s_1] + \dots + \\ &\quad [\ell(s_t, a_t) + \lambda_{t+1}^\top f(s_t, a_t) - \lambda_t^\top s_t] + \dots + [\ell(s_T, a_T) + \lambda_T^\top s_T]. \end{aligned}$$

2. Necessary condition for optimality: Total derivative of  $\mathcal{L}$  has to be zero:

$$\frac{\partial \mathcal{L}}{\partial \lambda_t} = 0, \quad \frac{\partial \mathcal{L}}{\partial s_t} = 0, \quad \frac{\partial \mathcal{L}}{\partial a_t} = 0.$$

# Gradients in single shooting (2)

$$\mathcal{L}(s, a, \lambda) = \sum_{t=0}^T \ell(s_t, a_t) + \sum_{t=0}^{T-1} \lambda_{t+1}^\top (f(s_t, a_t) - s_{t+1})$$

3. State equation:

$$\frac{\partial \mathcal{L}}{\partial \lambda_t} = s_{t+1} - f(s_t, a_t) \stackrel{!}{=} 0 \quad \Leftrightarrow \quad s_{t+1} = f(s_t, a_t) \quad \text{for } t = 0, \dots, T-1.$$

4. Adjoint equation:

$$\frac{\partial \mathcal{L}}{\partial s_t} = \frac{\partial \ell}{\partial s_t} + \left( \frac{\partial f}{\partial s_t} \right)^\top \lambda_{t+1} - \lambda_t \stackrel{!}{=} 0 \quad \Leftrightarrow \quad \lambda_t = \frac{\partial \ell}{\partial s_t} + \left( \frac{\partial f}{\partial s_t} \right)^\top \lambda_{t+1} \quad \text{for } t = 0, \dots, T-1.$$

5. Final adjoint state ( $t = T$ ): No influence on a future state,  $\lambda_T = \frac{\partial \ell}{\partial s_T}$ .

6. Action gradient:

$$\frac{\partial \mathcal{L}}{\partial a_t} = \frac{\partial \ell}{\partial a_t} + \left( \frac{\partial f}{\partial a_t} \right)^\top \lambda_{t+1} \quad \text{for } t = 0, \dots, T-1.$$



# Gradients in single shooting (3)

## Gradient calculation in the single shooting method for solving (1)

Given: Initial state  $s_0$ , action sequence  $a = (a_0, \dots, a_{T-1})$ .

- **Forward simulation** of the state equation:

$$s_{t+1} = f(s_t, a_t), \quad t = 0, 1, \dots, T-1.$$

- “Initial condition” (= condition at final time  $T$ ) for the adjoint equation:  $\lambda_T = \frac{\partial \ell}{\partial s_T}$ .
- **Backward simulation** of the adjoint equation:

$$\lambda_t = \frac{\partial \ell}{\partial s_t} + \left( \frac{\partial f}{\partial s_t} \right)^\top \lambda_{t+1}, \quad t = T-1, T-2, \dots, 1, 0.$$

- **Gradient calculation w.r.t.  $a$ :**

$$\frac{\partial \mathcal{L}}{\partial a_t} = \frac{\partial \ell}{\partial a_t} + \left( \frac{\partial f}{\partial a_t} \right)^\top \lambda_{t+1}, \quad t = 0, 1, \dots, T-1.$$

## Notes

- The backward simulation is closely related to the backward pass in backpropagation.
- In modern packages such as PyTorch, JAX or TensorFlow, this procedure can be automated with autodiff, when appropriately implementing the forward pass.

# Solving the OCP: full discretization

- Alternative to forward-backward: the full discretization
- Both  $s$  and  $a$  become optimization variables:

$$x = [s_0^\top, a_0^\top, s_1^\top, a_1^\top, \dots, a_{T-1}^\top, s_T^\top]^\top.$$

- Reformulation of the OCP (1)

$$\min_x F(x) = \sum_{t=0}^T \ell(s_t, a_t) \quad \text{s.t.} \quad C(x) = \begin{bmatrix} f(s_0, a_0) - s_1 \\ f(s_1, a_1) - s_2 \\ \vdots \\ f(s_{T-1}, a_{T-1}) - s_T \end{bmatrix} = 0 \quad (2)$$

- Total number of optimization variables when the state dimension is  $n$  and the action dimension is  $m$ :  $(T + 1)n + Tm$ .
- Solution via standard solvers for nonlinear, constrained optimization problems:
  - Require the gradients  $\nabla_x F(x)$  and  $\nabla_x C(x)$ .
  - The latter is a very large matrix, but it is fortunately very sparse.

# Comparison: Shooting vs. full discretization

|                                    | Shooting (1)                                                                                                                                                                                                    | Full discretization (2)                                                                                                                                                                                                                     |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sensitivity to initialization      | Very sensitive to poor initial guesses $a^{(0)}$ .                                                                                                                                                              | Because we initialize the entire state trajectory $s$ , the system doesn't have to be dynamically feasible during early solver iterations $\Rightarrow$ Significantly more stable for unstable dynamics.                                    |
| Computational cost vs. convergence | Sequential: Fast iterations (one forward and backward pass), but may take many iterations to converge.                                                                                                          | Full Discretization: Iterations computationally heavier, but standard <b>Sequential Quadratic Programming</b> (SQP) or <b>Interior Point</b> methods achieve quadratic convergence near the solution, requiring far fewer total iterations. |
| Constraint Handling                | Limits on states like $s_{\min} \leq s_t \leq s_{\max}$ or actuations ( $a_{\min} \leq a_t \leq a_{\max}$ ) can be handled easily. They are simply treated as bound constraints on elements of our vector $x$ . | Handling state constraints requires challenging penalty functions or barrier methods.                                                                                                                                                       |

# The linear-quadratic case

# The linear-quadratic case

Let's consider a special (yet very common) case

- **Linear dynamics:**

$$s_{t+1} = As_t + Ba_t, \quad \text{with } A \in \mathbb{R}^{n \times n} \text{ and } B \in \mathbb{R}^{n \times m}.$$

- **Quadratic losses:**

$$L(s, a) = \left( \sum_{t=0}^{T-1} s_t^\top Q s_t + a_t^\top R a_t \right) + s_T^\top Q_f s_T,$$

with matrices  $Q, Q_f \in \mathbb{R}^{n \times n}$  penalizing the (terminal) state and  $R \in \mathbb{R}^{m \times m}$  penalizing the control cost.

## Additional conditions

- The matrix  $Q$  is *positive semidefinite*:

$$s^\top Q s \geq 0 \quad \forall s \in \mathbb{R}^n.$$

- The matrix  $R$  is *positive definite*:

$$a^\top R a \geq 0 \quad \forall a \in \mathbb{R}^m.$$

- These ensure existence and uniqueness of a minimizer  $s^*, a^*$ .

## Linear-quadratic optimal control problem

$$\begin{aligned} & \min_{s,a} \left( \sum_{t=0}^{T-1} s_t^\top Q s_t + a_t^\top R a_t \right) + s_T^\top Q_f s_T \\ \text{subject to } & s_{t+1} = A s_t + B a_t, \quad t = 0, 1, \dots, T-1. \end{aligned} \tag{3}$$

# Removing the constraints

Let's explicitly calculate  $s_1, s_2, \dots$ :

$$s_1 = As_0 + Ba_0$$

$$s_2 = As_1 + Ba_1 = A(As_0 + Ba_0) + Ba_1 = A^2s_0 + ABa_0 + Ba_1$$

$$s_3 = As_2 + Ba_2 = A(A^2s_0 + ABa_0 + Ba_1) + Ba_2 = A^3s_0 + A^2Ba_0 + ABa_1 + Ba_2$$

General rule:  $s_t = A^t s_0 + \sum_{i=0}^t A^{t-i-1} B a_i$

Explicitly:

$$\underbrace{\begin{bmatrix} s_0 \\ s_1 \\ s_2 \\ s_3 \\ \vdots \\ s_T \end{bmatrix}}_{\hat{s} \in \mathbb{R}^{(T+1)n}} = \underbrace{\begin{bmatrix} 0 & 0 & 0 & \dots & B \\ 0 & 0 & 0 & \dots & 0 \\ AB & B & 0 & \dots & 0 \\ A^2B & AB & B & \ddots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ A^{T-1}B & A^{T-2}B & A^{T-3}B & \dots & B \end{bmatrix}}_{G \in \mathbb{R}^{(T+1)n \times Tm}} \underbrace{\begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \\ \vdots \\ a_{T-1} \end{bmatrix}}_{\hat{a} \in \mathbb{R}^{Tm}} + \underbrace{\begin{bmatrix} I \\ A \\ A^2 \\ A^3 \\ \vdots \\ A^T \end{bmatrix}}_{H \in \mathbb{R}^{(T+1)n}} s_0$$

In short:  $\hat{s} = G\hat{a} + Hs_0$

# Reformulating the cost function

- Next, we stack our penalty matrices such that they match the dimensions of  $S$  and  $U$ :

$$\hat{Q} = \begin{bmatrix} Q & & \\ & \ddots & \\ & & Q \\ & & & Q_f \end{bmatrix} \in \mathbb{R}^{(T+1)n \times (T+1)n}, \quad \hat{R} = \begin{bmatrix} R & & \\ & \ddots & \\ & & R \end{bmatrix} \in \mathbb{R}^{Tm \times Tm}.$$

- We can now insert this into the linear-quadratic OCP (3):

$$\min_{s,a} \left( \sum_{t=0}^{T-1} s_t^\top Q s_t + a_t^\top R a_t \right) + s_T^\top Q_f s_T \quad \text{s.t.} \quad s_{t+1} = A s_t + B a_t, \quad t = 0, 1, \dots, T-1$$

becomes

$$\min_{\hat{s}, \hat{a}} \hat{s}^\top \hat{Q} \hat{s} + \hat{a}^\top \hat{R} \hat{a} \quad \text{s.t.} \quad \hat{s} = G \hat{a} + H s_0.$$



- **Question:** Do we have to consider  $\hat{s}$  and  $\hat{a}$  independently?  
⇒ Let's simply insert the constraints!

$$\min_{\hat{a}} (G\hat{a} + Hs_0)^\top \hat{Q} (G\hat{a} + Hs_0) + \hat{a}^\top \hat{R} \hat{a} \quad (4)$$

# Solving the linear OCP

$$\begin{aligned}
 & \min_{\hat{a}} (G\hat{a} + Hs_0)^\top \hat{Q} (G\hat{a} + Hs_0) + \hat{a}^\top \hat{R} \hat{a} \\
 &= \min_{\hat{a}} (\hat{a}^\top G^\top + s_0^\top H^\top) \hat{Q} (G\hat{a} + Hs_0) + \hat{a}^\top \hat{R} \hat{a} \\
 &= \min_{\hat{a}} \hat{a}^\top G^\top \hat{Q} G \hat{a} + 2\hat{a}^\top G^\top \hat{Q} H s_0 + s_0^\top H^\top \hat{Q} H s_0 + \hat{a}^\top \hat{R} \hat{a} \\
 &= \min_{\hat{a}} \hat{a}^\top (G^\top \hat{Q} G + \hat{R}) \hat{a} + 2\hat{a}^\top G^\top \hat{Q} H s_0 + s_0^\top H^\top \hat{Q} H s_0
 \end{aligned}$$

Some linear algebra

$$\begin{aligned}
 (G\hat{a})^\top &= \hat{a}^\top G^\top \\
 s_0^\top H^\top \hat{Q} G \hat{a} &= \hat{a}^\top G^\top \hat{Q} H s_0
 \end{aligned}$$

- This is a quadratic polynomial in  $\hat{a}$ !
- Since  $Q$  and  $R$  (and thus,  $\hat{Q}$  and  $\hat{R}$ ) are positive (semi-)definite, the loss function is strictly convex  $\Rightarrow$  unique minimizer.
- Take the derivative with respect to  $\hat{a}$ :

$$\frac{d}{d\hat{a}} \left( \hat{a}^\top (G^\top \hat{Q} G + \hat{R}) \hat{a} + 2\hat{a}^\top G^\top \hat{Q} H s_0 + s_0^\top H^\top \hat{Q} H s_0 \right) = 2(G^\top \hat{Q} G + \hat{R}) \hat{a} + 2G^\top \hat{Q} H s_0.$$

- Setting the gradient to zero yields a linear system (depending on the initial condition  $s_0$ ):

$$\boxed{\left(G^{\top} \hat{Q} G + \hat{R}\right) \hat{a} = -G^{\top} \hat{Q} H s_0.} \quad (5)$$

# Example: Control of a forklift (1)

- Consider a simple vehicle driving on a plane (with mass  $m$  and friction  $d$ ):

$$s = \begin{bmatrix} p_1 \\ v_1 \\ p_2 \\ v_2 \end{bmatrix}, \quad \dot{s} = \frac{ds}{dt}(t) = A_c s(t) + B_c a(t) = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & -d/m & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & -d/m \end{bmatrix} s(t) + \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} a(t).$$



- This can be translated into a discrete-time system with time step  $\Delta t$ :

$$A = \exp(A_c \Delta t), \quad B = \left( \int_0^{\Delta t} \exp(A(\Delta t - \tau)) d\tau \right) B_c \quad \Rightarrow \quad s_{t+1} = A s_t + B a_t.$$

- Control objective: Steer the forklift from  $s_0 = [2 \ 0 \ 0 \ 0]^T$  to the origin with a small penalty on the control:

$$\min_{s,a} \sum_{t=0}^T \|s_t\|_2^2 + \alpha \|a_t\|_2^2 \quad \text{s.t.} \quad s_{t+1} = A s_t + B a_t.$$

## **Example: Control of a forklift (2)**

- Define individual control matrices:

$$Q = \begin{bmatrix} 1 & & & \\ & 0 & & \\ & & 1 & \\ & & & 0 \end{bmatrix}, \quad Q_f = 10 \cdot Q, \quad R = \begin{bmatrix} 0.01 & \\ & 0.01 \end{bmatrix}.$$

- Assemble the matrices  $\hat{Q}$  and  $\hat{R}$ . With  $T = 200$ , we get

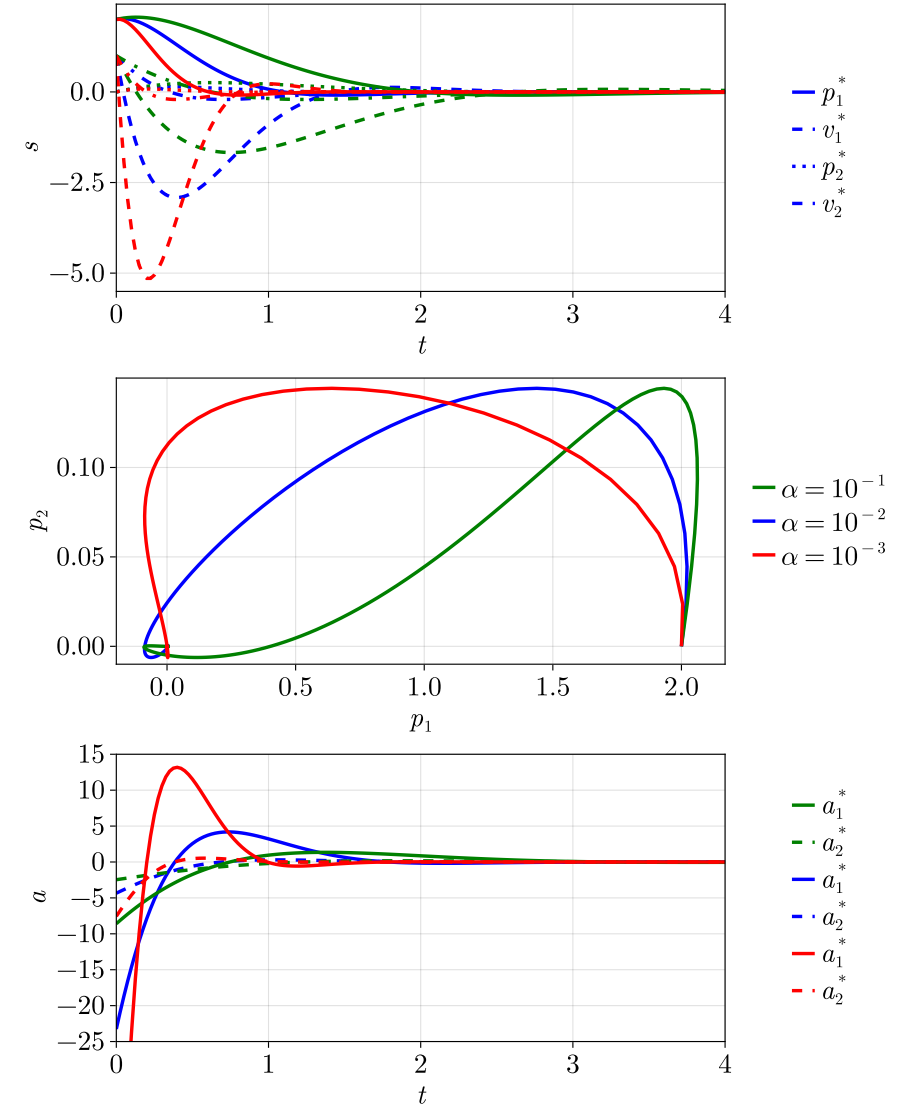
$$\hat{Q} \in \mathbb{R}^{804 \times 804} \quad \text{and} \quad \hat{R} \in \mathbb{R}^{400 \times 400}.$$

- Given the initial condition  $s_0$ , solve the linear system

$$(G^\top \hat{Q} G + \hat{R}) \hat{a} = -G^\top \hat{Q} H s_0. \quad (5)$$

- Simulate dynamics to get optimal trajectory  $s^*$ :

$$s_{t+1}^* = A s_t^* + B a_t^* \quad \text{or} \quad \hat{s}^* = G \hat{a}^* + H s_0.$$



# Closed-loop control

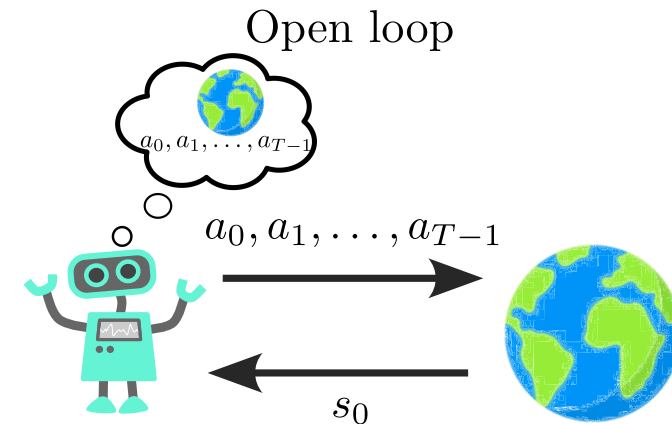
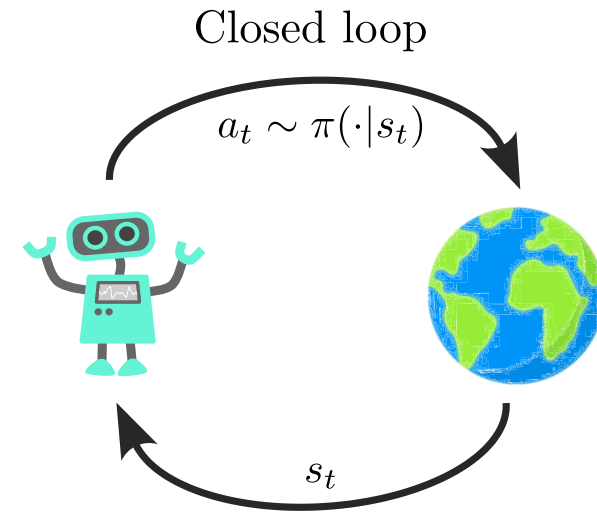
# Model predictive control (MPC)



- Given a model, solving open-loop OCPs is a well-established field.
- But there are also **drawbacks** in the application to real systems.
  - Unforeseen events cannot be accounted for.
  - External disturbances/uncertainties are hard/impossible to include.
  - Smalles inaccuracies scale exponentially over time.
- **Question:** Can we close the loop around OCPs?

### Model predictive control (MPC)

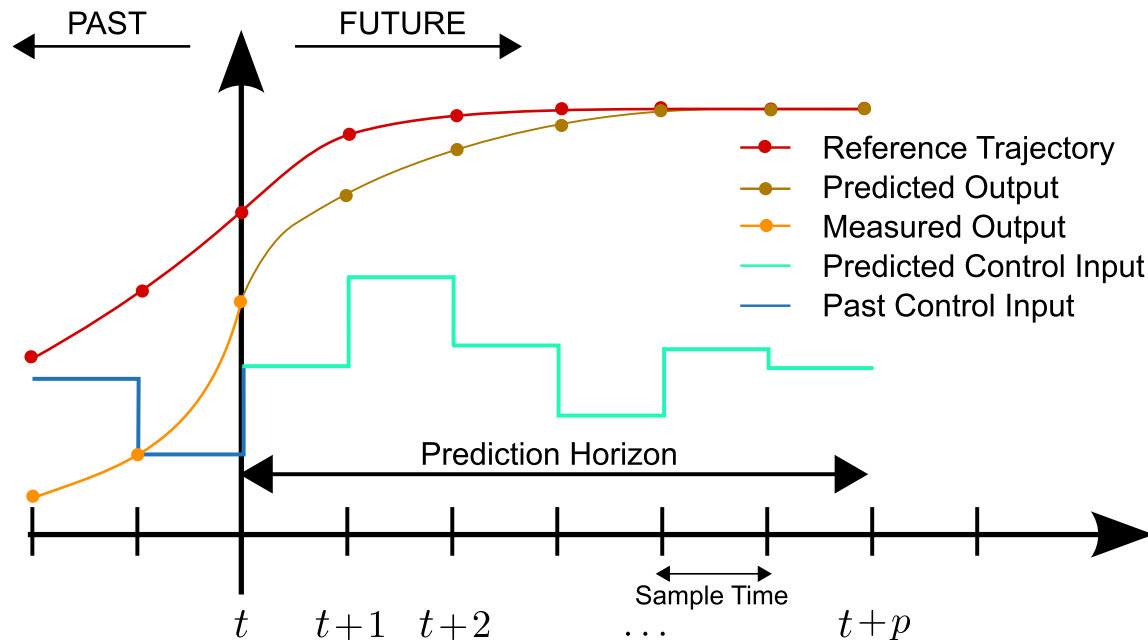
- Let us solve the open-loop problem repeatedly on a shorter horizon  $p \leq T$ .
- The “feedback” is setting the initial condition of the OCP.



Only measure at  $s_0$ , no feedback

*Inspired by Sergey Levine's CS285 lecture.*

# MPC scheme



Source: Wikipedia.

## The MPC loop

1. Measure system state  $s_t$  and use it as initial state for the OCP (1).
2. Solve (1) using a model of the real system over a prediction horizon of length  $p$ .
3. Apply first entry of  $a^*$  (i.e.,  $a_0^*$ ) to the real system.
4. Advance real system by one time step.
5. Repeat.

## Pros and cons

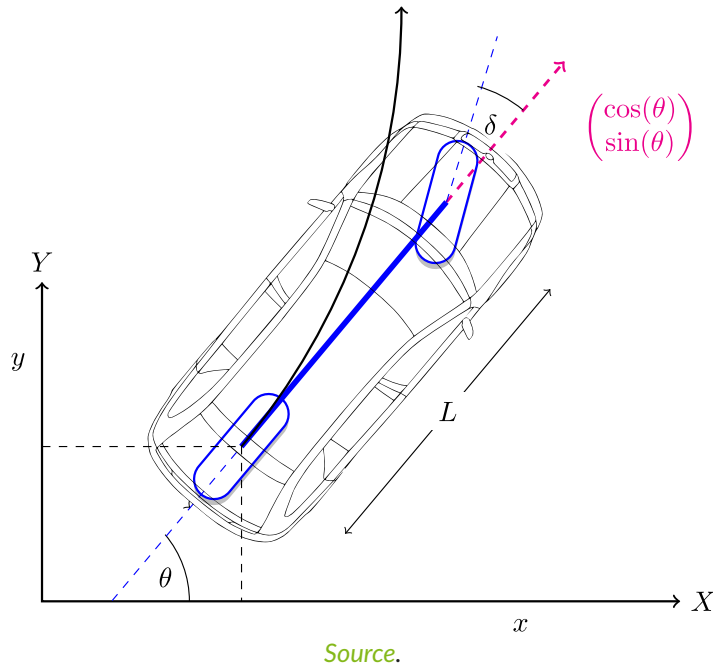
- + Very robust.
- + No offline learning phase (if model known).
- + Easy to include state constraints.
- + Lots of theory. (e.g., (Grüne and Pannek 2017))

- Real-time capability (in particular for nonlinear models).
- Control performance depends on model quality.

# Example: Autonomous driving

We want to plan the trajectory of an autonomous vehicle from an initial position towards a terminal position.

- Dynamics: Simplified **kinematic bicycle model** (Kong et al. 2015).
- MPC optimization: Adam ( $\eta = 0.1$ ); 10 steps.



$$s = \begin{bmatrix} x \\ y \\ \theta \\ v \end{bmatrix}, \quad a = \begin{bmatrix} \text{acc} \\ \delta \end{bmatrix},$$

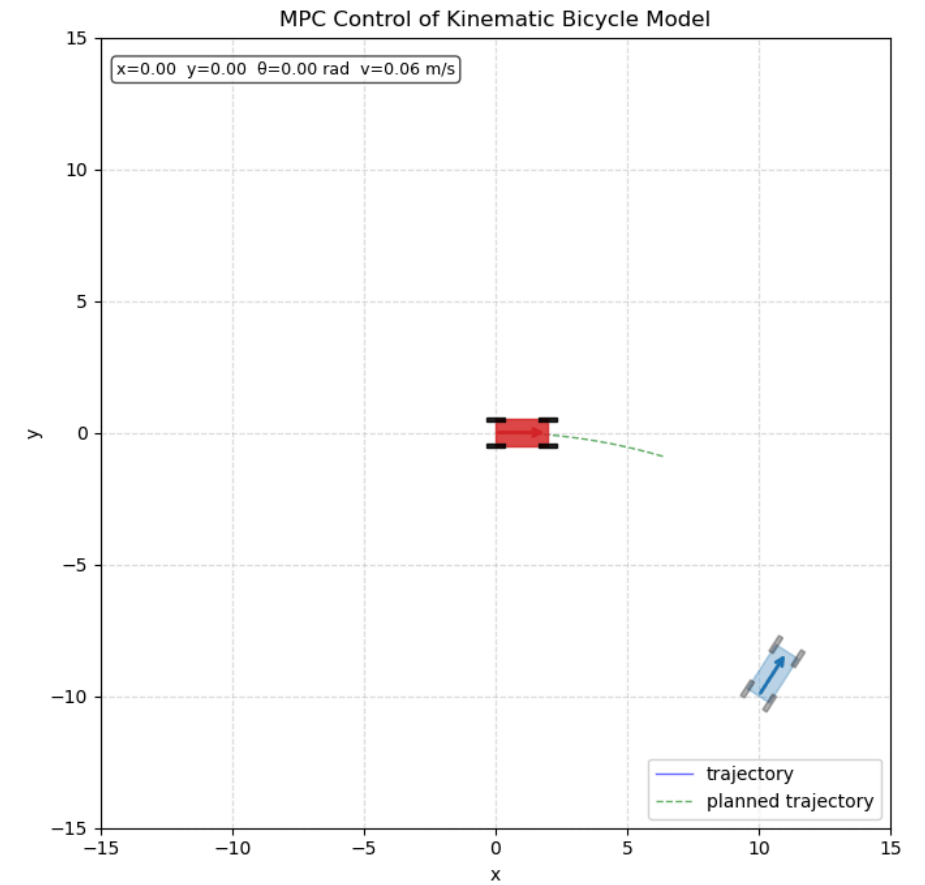
$$\dot{s} = \begin{bmatrix} s_4 \cos s_3 \\ s_4 \sin s_3 \\ \frac{s_4}{L} \tan a_2 \\ a_1 - \lambda s_4 \end{bmatrix}.$$

- Objective: **terminal condition**

$$\min_{a_1, \dots, a_p} \ell(s_p) = \|s_p - s_p^{\text{target}}\|_2^2 \quad \text{s.t.} \quad \text{bicycle model.}$$

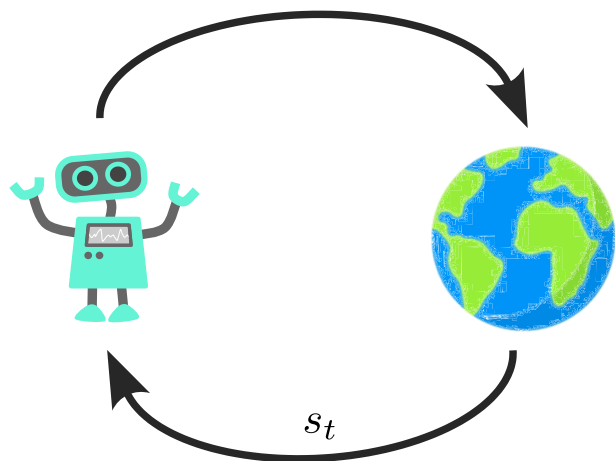
- Discretization:  $\Delta t = 0.2$ ,  $p = 25$ .

- Results:



# Relation to reinforcement learning

## Reinforcement Learning



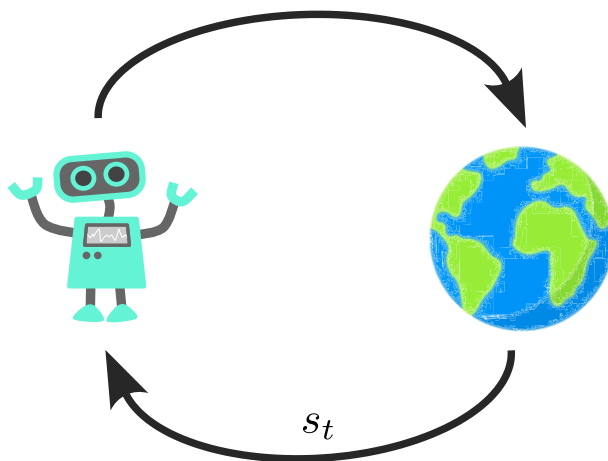
$$a_t \sim \pi(\cdot | s_t)$$

VS.

## Model Predictive Control

$$\begin{aligned} a_t = \arg \min_{\tilde{a}} \sum_{\tau=0}^p \ell(\tilde{s}_{\tau}, \tilde{a}_{\tau}) \\ \text{s.t. } \tilde{s}_{\tau+1} = f(\tilde{s}_{\tau}, \tilde{a}_{\tau}) \\ \tilde{s}_0 = s_t \end{aligned}$$

VS.



## Trade-offs

- Learning a policy offline vs. solving an OCP online.
- Expensive training vs. expensive inference.

## Approaches in between

- Explicit MPC
- Differentiable predictive control (DPC)
- Monte-Carlo tree search (MCTS)

# Explicit MPC

- MPC's main challenge: **large online cost!**
- Can we avoid this?  $\Rightarrow$  Let's revisit the OCP formulation (1) in the MPC setting:

$$\begin{aligned} \min_{\tilde{a}} \quad & \sum_{\tau=0}^p \ell(\tilde{s}_{\tau}, \tilde{a}_{\tau}) \\ \text{s.t.} \quad & \tilde{s}_{\tau+1} = f(\tilde{s}_{\tau}, \tilde{a}_{\tau}), \quad \tilde{s}_0 = s_t. \end{aligned} \quad (6)$$

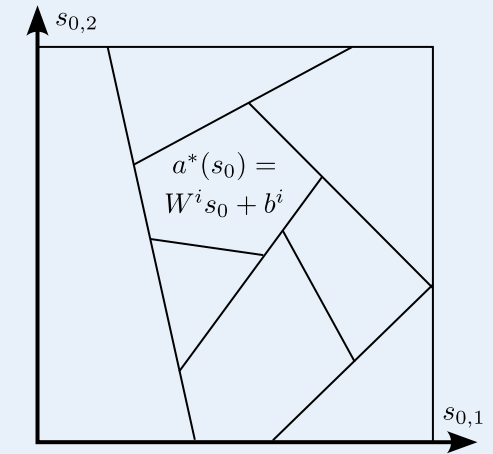
- This OCP is parameterized by the initial condition  $s_0$ !
- If we solve it for every  $s_0 \in \mathcal{S}$  and explicitly store  $a^*(s_0)$  in a library, we can quickly retrieve it in the online phase.
- But there are infinitely many  $s_0$ ! 🤖
- Options:
  - Approximate by a finite number of  $s_0$ .

The explicit linear-quadratic regulator (Bemporad et al. 2002)

- For linear models and quadratic costs,

$$\begin{aligned} f(s, a) &= As + Ba, \\ \ell(s, a) &= s^{\top} Q s + a^{\top} R a, \end{aligned}$$

the solution space  $a^*(s_0)$  for (6) is split into polygons on which the optimal feedback  $a^*$  is an affine function of the initial condition  $s_0$



- Finite number of polygons  $\Rightarrow$  finite number of OCPs to solve!
- Algorithm for the **offline phase**:
  1. Split the space of initial conditions into polygons  $i = 1, \dots, N_p$ .
  2. Determine coefficients  $W^i$  and  $b_i$  for the affine feedback law (details in (Bemporad et al. 2002)).
- Drawback: Exponential growth of polygons with state dimension  $n$ .
  - Workaround: Replace polygon by deep neural net (Karg and Lucia 2020).

# Differentiable predictive control (DPC)

- Let's start with a policy network, just as in policy gradient methods:  $a = \mu_\phi(s)$ .
- Inserting into the right-hand side of our model yields an autonomous system:

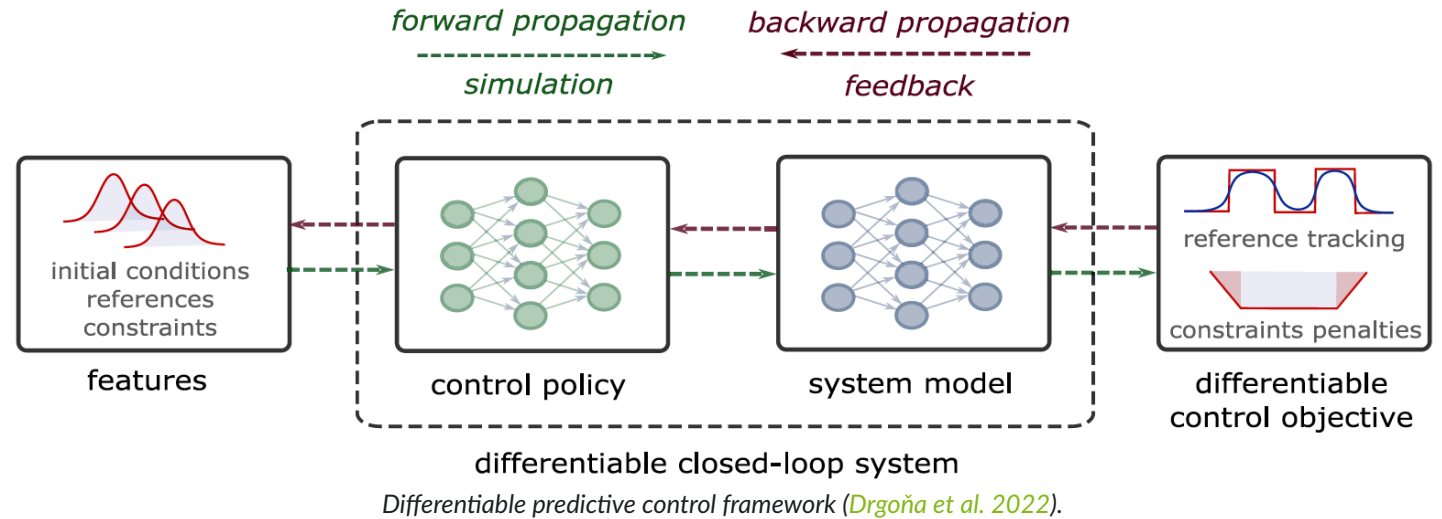
$$s_{t+1} = f(s_t, \mu_\phi(s_t)).$$

- We can now formulate a closed-loop performance criterion:

$$L(\phi) = \sum_{t=1}^T \|s_t - s_t^{\text{target}}\| + \alpha \|\mu_\phi(s_t)\|.$$

- If the dynamics model  $f$  (or a learned version  $f_\theta$ ; see next lecture) is differentiable, then we can directly optimize the policy parameter  $\phi$  via backpropagation and gradient descent:

$$\phi \leftarrow \phi - \eta \frac{dL}{d\phi}.$$



- This is known as **differentiable predictive control (DPC)**.



# Summary / what we have learned

- Knowing a model of the system dynamics can be very useful, not only for tabular reinforcement learning methods.
- Open-loop = Planning: Given an initial condition  $s_0$ , find the optimal actions for an entire trajectory **once**.
- MPC: Repeated solution of open-loop optimal control problems.
- Explicit MPC: Solving the open-loop problem in an offline phase in a parameterized fashion.
- Differentiable predictive control (DPC): Policy optimization via direct differentiation through the dynamics model.

# References

- Bemporad, Alberto, Manfred Morari, Vivek Dua, and Efstratios N. Pistikopoulos. 2002. “The Explicit Linear Quadratic Regulator for Constrained Systems.” *Automatica* 38 (1): 3–20. doi:[https://doi.org/10.1016/S0005-1098\(01\)00174-1](https://doi.org/10.1016/S0005-1098(01)00174-1).
- Drgoňa, Ján, Karol Kiš, Aaron Tuor, Draguna Vrabie, and Martin Klaučo. 2022. “Differentiable Predictive Control: Deep Learning Alternative to Explicit Model Predictive Control for Unknown Nonlinear Systems.” *Journal of Process Control* 116: 80–92.
- Grüne, L., and J. Pannek. 2017. *Nonlinear Model Predictive Control*. 2nd ed. Springer International Publishing.
- Karg, Benjamin, and Sergio Lucia. 2020. “Efficient Representation and Approximation of Model Predictive Control Laws via Deep Learning.” *IEEE Transactions on Cybernetics* 50 (9): 3866–78. doi:[10.1109/TCYB.2020.2999556](https://doi.org/10.1109/TCYB.2020.2999556).
- Kong, Jason, Mark Pfeiffer, Georg Schildbach, and Francesco Borrelli. 2015. “Kinematic and Dynamic Vehicle Models for Autonomous Driving Control Design.” In *2015 IEEE Intelligent Vehicles Symposium (IV)*, 1094–99. doi:[10.1109/IVS.2015.7225830](https://doi.org/10.1109/IVS.2015.7225830).